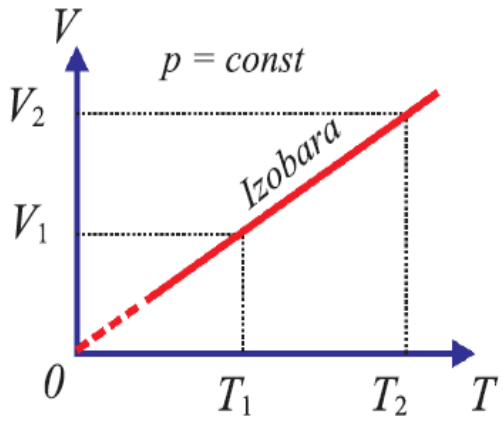
Fizika

9 sinf

MAVZU: Izobarik jarayon



Izobarik jarayon

Ideal gazning massasi (m=const) va bosimi (p=const) o‘zgarmas bo‘lgandagi gaz holatining o‘zgarish jarayoniga ***izobarik jarayon*** deyiladi.

Grekcha “baros” bosim degan ma’noni anglatadi.

Izobarik jarayonda o‘zgarmas massali gazning hajmi (V) temperaturasi (T) ga bog‘liq ravishda o‘zgaradi.

Izobarik jarayon

p =const $p_1 = p_2$

$$p_1 V_1 = \frac{m}{M} \cdot R \cdot T_1 \qquad p_2 V_2 = \frac{m}{M} \cdot R \cdot T_2$$

Bu tenglamalarni hadma-had bo‘lib, quyidagini hosil qilamiz:

$$\frac{V_1}{V_2} = \frac{T_1}{T_2} \qquad \text{yoki} \qquad \frac{V_1}{T_1} = \frac{V_2}{T_2}$$

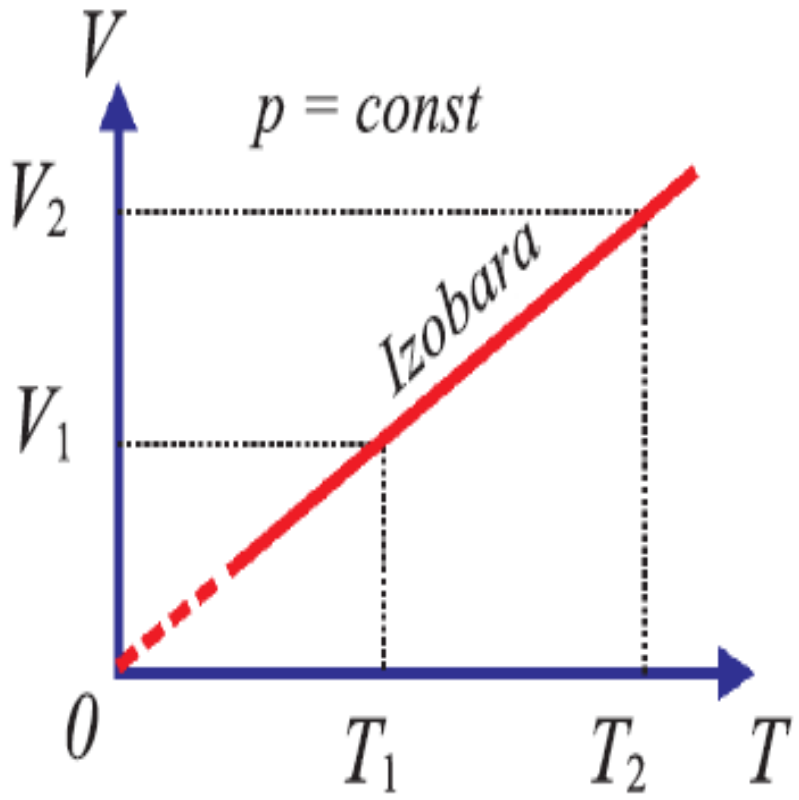
$\frac{V}{T}$ =const , Gey-Lyussak qonuni (1802-yil)

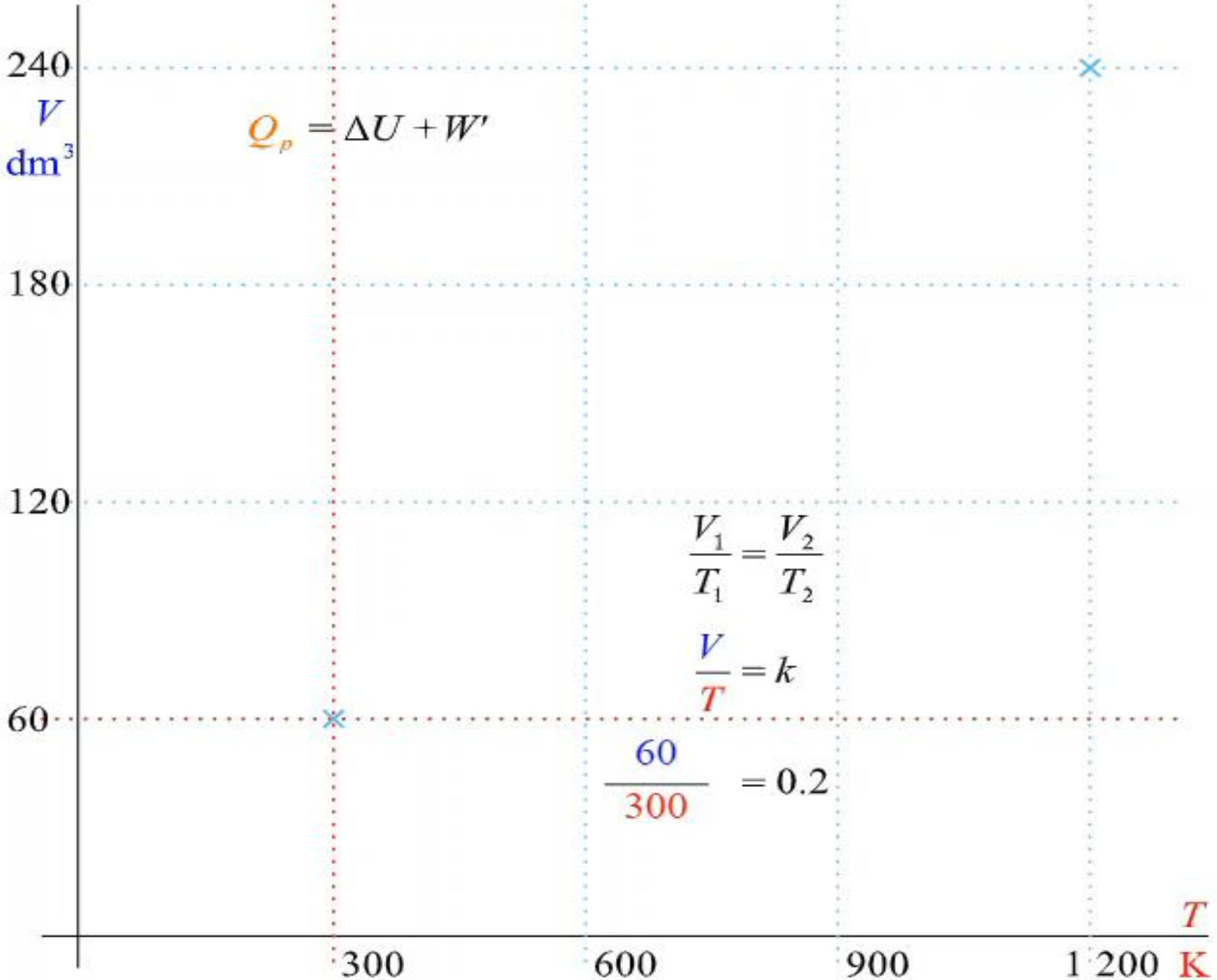
Izobarik jarayon grafigi

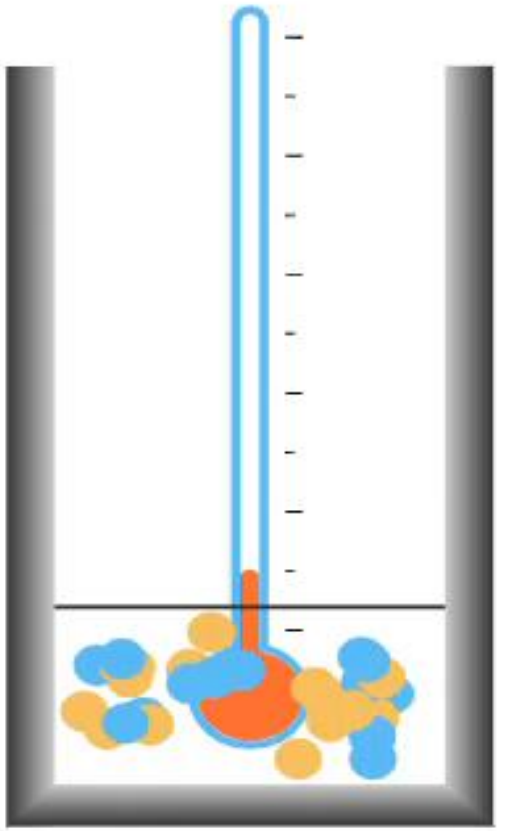
$$\frac{V}{T} = \text{const}$$

Izobarik jarayonda berilgan gazning hajmi bilan temperaturasi orasidagi munosabatni ifodalovchi chiziq ***izobara chizig‘i*** deyiladi.

O‘zgarmas bosimda berilgan massali gazning hajmi tempera-turaga to‘g‘ri proporsional ravishda o‘zgaradi.




240
V
dm3
180
120
60
Qp = ΔU + W'
V1/T1 = V2/T2
V/T = k
60/300 = 0.2
T
300
600
900
1 200
K





1-namunaviy masala

Ideal gazning temperaturasi $67℃$ va hajmi 25 l.

Bosim o‘zgarmaganda hajm 10 l ga teng bo‘lishi uchun gazni qanchaga sovitish kerak?

Berilgan:

$T_1 = 67\ ^\circ\text{C} + 273 = 340\ \text{K}$
$V_1 = 25\ l = 25 \cdot 10^{-3}\ \text{m}^3$
$V_2 = 10\ l = 10 \cdot 10^{-3}\ \text{m}^3$
$p = const.$

Topish kerak:
$\Delta T = ?$

Formulasi:

$$\frac{V_1}{V_2} = \frac{T_1}{T_2};\quad T_2 = \frac{V_2 \cdot T_1}{V_1};$$

$$\Delta T = T_1 - T_2.$$

$$[\Delta T] = \text{K}.$$

Hisoblash:

$$T_2 = \frac{10 \cdot 10^{-3} \cdot 340\ \text{K}}{25 \cdot 10^{-3}} = 136\,\text{K}.$$

$$\Delta T = 340\ \text{K} - 136\ \text{K} = 204\ \text{K}.$$

Javob: $\Delta T = 204\ \text{K}$.

9-mashq

1 - masala

Temperaturasi 27℃ bo‘lgan ideal gazning hajmi 10 l edi. Gazni izobarik ravishda 327℃ gacha isitilganda uning hajmi qanday o‘zgaradi.

Berilgan:

$t_1 = 27℃$

$V_1 = 10\ l$

$t_2 = 327℃$

$p = const.$

Topish kerak:

$\Delta V - ?$

Formula:

$T = t + 273$

$\Delta V = V_2 - V_1$

$p = const$

$$\frac{V_1}{V_2} = \frac{T_1}{T_2}$$

$$V_2 = \frac{T_2 \cdot V_1}{T_1}$$

Yechish:

$T_1 = 27℃ + 273 = 300\ K$

$T_2 = 327℃ + 273 = 600\ K$

$$V_2 = \frac{600\ K \cdot 10\ l}{300\ K} = 20\ l$$

$\Delta V = 20\ l - 10\ l = 10\ l$

Javob: 10 l ga ortdi.

9-mashq

4 - masala

Ideal gaz 47℃ da 3 l hajmni egallagan. Bosimni o‘zgartirmasdan hajmni 1,2 l ga orttirish uchun gazning temperaturasini qanchaga ko‘tarish kerak?

Berilgan:

$t_1 = 47℃$

$V_1 = 3\ l$

$\Delta V = 1{,}2\ l$

$p = const.$

T.K.: $\Delta t - ?$

Formula:

$T = t + 273$

$\Delta t = \Delta T$

$\Delta T = T_2 - T_1$

$\frac{V_1}{V_2} = \frac{T_1}{T_2}$

$T_2 = \frac{V_2 \cdot T_1}{V_1}$

$\Delta V = V_2 - V_1$

$V_2 = V_1 + \Delta V$

Yechish:

$T_1 = 47℃ + 273 = 320\ K$

$V_2 = 3\ l + 1{,}2\ l = 4{,}2\ l$

$T_2 = \frac{4{,}2\ l \cdot 320\ K}{3\ l} = 448\ K$

$\Delta T = 448\ K - 320\ K = 128\ K$

$\Delta t = \Delta T = 128℃$

Javob: 128℃ ga ko‘tarish kerak

Mustaqil bajarish uchun topshiriqlar

1. 36-betdagi mavzuga doir savollarga javob yozish.

2. Ideal gazning tempe-raturasi 51°Cva hajmi 0,9 l. Bosim o‘zgarmaganda hajm 0,3 l ga teng bo‘lishi uchun gazni qancha (K) sovitish kerak? (9-mashq 2-masala)

3. Gaz 27°C tempera-turada 3 l hajmga ega. Gaz izobarik 100°C ga qizdirilsa, qanday hajmni egallaydi? (9-mashq 3-masala)